Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 1 shows a prize that is made from a metal star on a plastic base.

Figure 1

A person starts to clean the prize by rubbing the plastic base with a dry cloth.

The plastic base becomes positively charged and the cloth becomes negatively charged.

(i) The plastic base has

(1)

- ☐ **A** gained electrons
- ☐ **B** gained protons
- ☐ **C** lost electrons
- ☐ **D** lost protons

(ii) Explain why the cloth has become negatively charged.

(2)

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(Total for question = 3 marks)

Q2.

A battery sends a current through a metal wire.
Plastic is an insulator.
A student rubs a piece of plastic with a cloth.
This gives the plastic a negative charge.
(i) Explain how the plastic is charged by the rubbing.

(2)

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(ii) The cloth is also charged when it rubs against the plastic.

Describe the charge on the cloth.

(2)

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Q3.

A student rubs a plastic comb with a dry cloth to give the comb a positive electric charge.

Figure 6 shows the charged plastic comb picking up small pieces of paper.

(Source © GIPhotoStock/SCIENCE PHOTO LIBRARY)

Figure 6

(i) Explain how rubbing the comb with a dry cloth gives the comb a positive electric charge.

(3)

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(ii) Explain how the positively-charged plastic comb picks up the small pieces of paper.

(3)

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(Total for question = 6 marks)

Q4.

Glass is an insulator.

A student rubs a piece of glass with some silk.

The glass becomes positively charged.

(i) Explain how rubbing silk charges the glass.

(2)

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(ii) The silk is also charged when it rubs against the glass.
Explain how the silk becomes charged.

(2)

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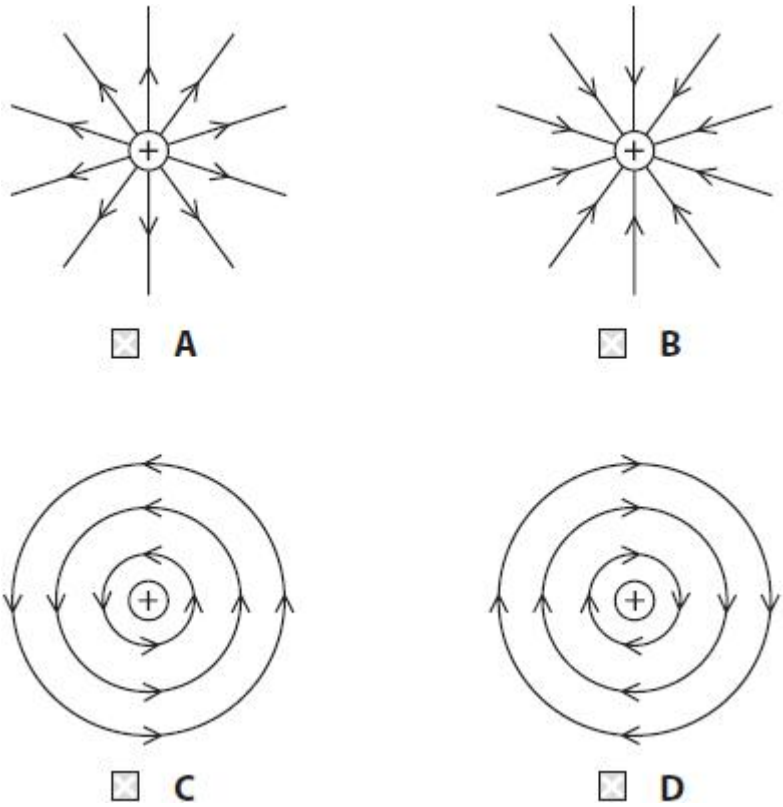
(Total for question = 4 marks)

Q5.

Which of these diagrams shows the shape and direction of the electric field around a positive point charge?

(1)



(Total for question = 1 mark)

Q6.

This question is about static electricity and electric fields.

Figure 2 shows the electric field around a point charge.

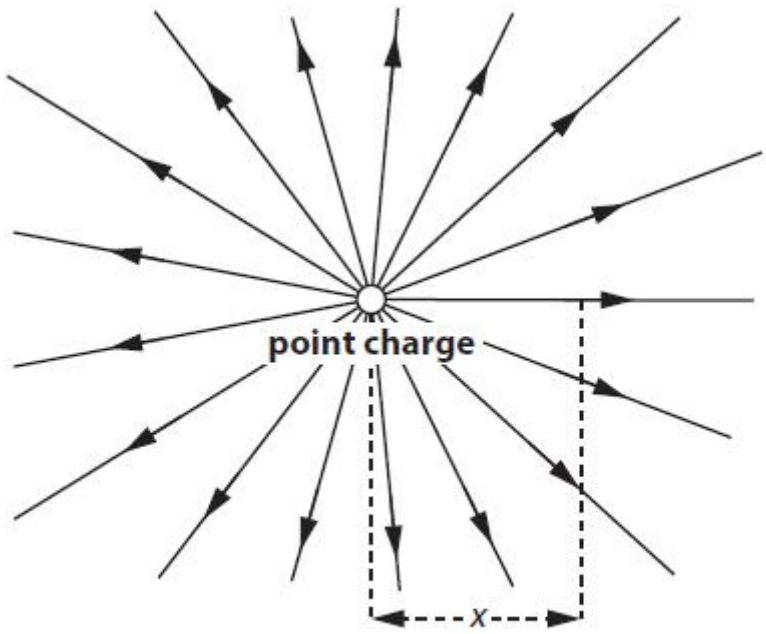


Figure 2

(i) State how Figure 2 gives evidence that the point charge is positive.

(1)

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(ii) In Figure 2, *x* is the distance from the point charge.

State how Figure 2 gives evidence that the electric field strength decreases as *x* increases.

(1)

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(iii) Figure 3 shows a graph of electric field strength against distance *x*.

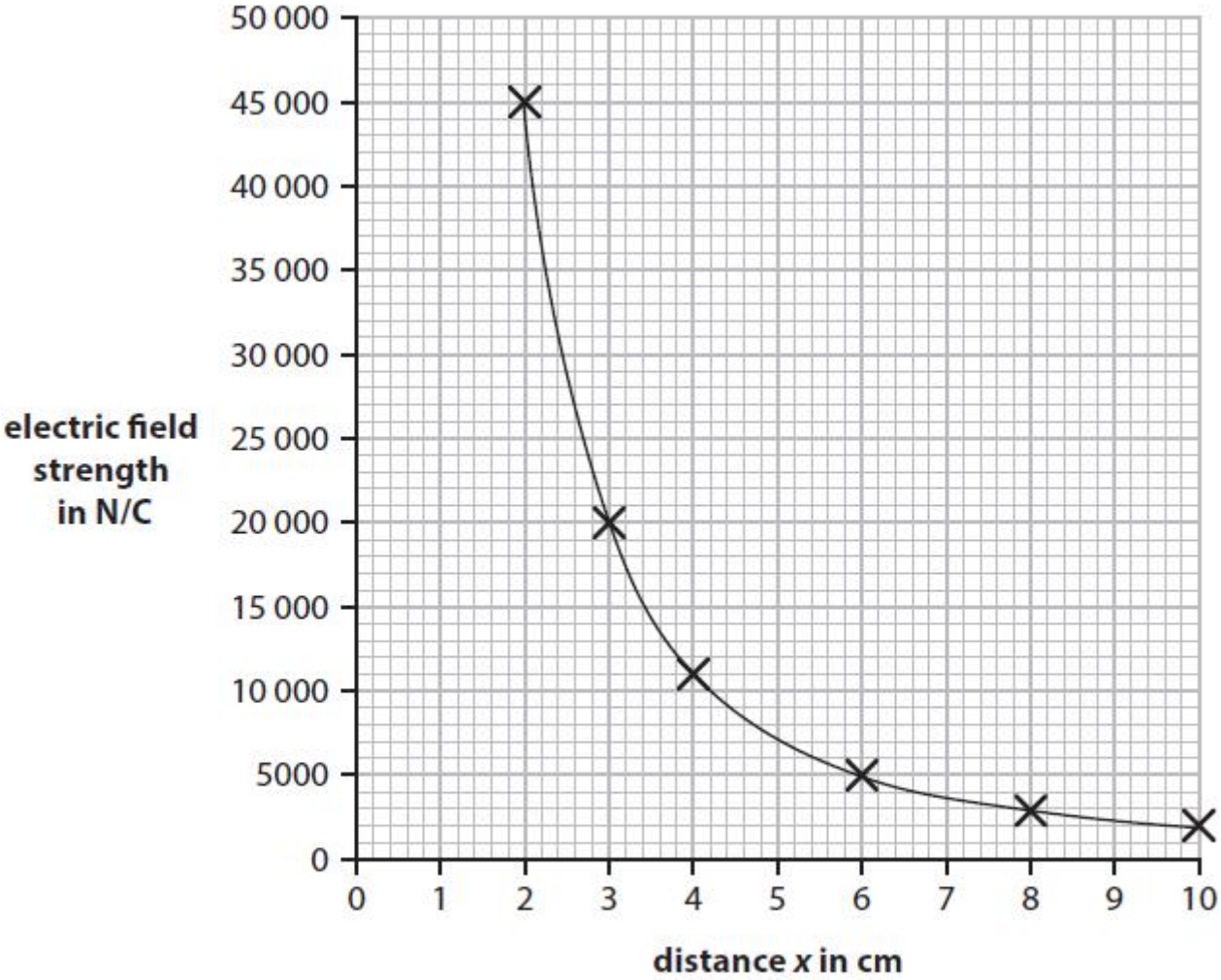

Figure 3

The equation relating electric field strength to distance, *x*, is

electric field strength $= \frac{A}{x^2}$

A is a constant.

Using data from the graph in Figure 3, calculate the value of A when *x* = 3 cm.

(2)

A = ... N cm^2/C

(Total for question = 4 marks)

Q7.

This question is about static electricity and electric fields.

Give the meaning of the term **electric field**.

(1)

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(Total for question = 1 mark)

Q8.

Figure 8 shows an airport worker refuelling an aircraft.

(Source: © Stanisław Tokarski/123RF)

Figure 8

Pumping fuel into an aircraft can be dangerous.

The worker connects an earth wire to the aircraft before pumping fuel.

Give **one** reason why earthing reduces the risk of fire.

(1)

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(Total for question = 1 mark)

Q9.

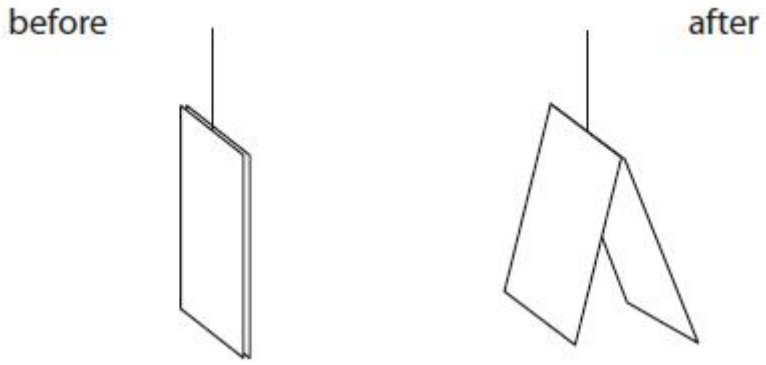
(a) A student rubs a plastic rod with a dry cloth. The cloth becomes positively charged.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.
The cloth becomes positively charged because

(1)

☐ **A** negative charge has moved from the cloth to the rod
☐ **B** negative charge has moved from the rod to the cloth
☐ **C** positive charge has moved from the cloth to the rod
☐ **D** positive charge has moved from the rod to the cloth.

(ii) Two plastic strips are joined at the top and are hanging by a thread.
The student rubs both strips with another dry cloth.
The diagram shows the two plastic strips before and after the student rubs them.



Explain why the strips behave in this way after they are rubbed with the cloth.

(2)

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(b) The student reads an article about the possible build-up of static electricity during the refuelling of an aircraft.

(i) Explain why this build-up could be dangerous.

(2)

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(ii) The article also states that the aircraft is connected by a metal cable to the ground. The aircraft is also connected by a metal cable to the refuelling tanker.
Explain how these cables reduce the dangers when refuelling the aircraft.

(3)

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(Total for Question = 8 marks)

Q10.

(i) Describe **one** situation where separation of electric charge can create a spark.

(2)

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(ii) In a spark, the total charge of 0.22 μC (microcoulombs) flows in 2 ms (milliseconds).
Calculate the average current in that time.

(4)

average current = ... A

(Total for question = 6 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 1 shows a paint sprayer.

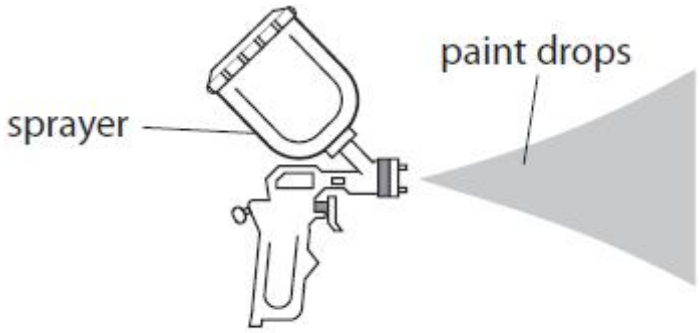

Figure 1

Some paint sprayers give the paint drops a positive charge as they leave the sprayer.

(i) The paint drops have a positive charge because the sprayer

(1)

- ☐ **A** removes electrons from the paint drops
- ☐ **B** adds electrons to the paint drops
- ☐ **C** removes protons from the paint drops
- ☐ **D** adds protons to the paint drops

(ii) Figure 2 shows the spray pattern from two different paint sprayers.

Figure 2

Sprayer X does not charge the paint drops. Sprayer Y gives the paint drops a positive charge.
Explain how charging the paint drops changes the shape of the spray pattern.

(2)

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(iii) Sprayer Y is used in a factory to paint a metal object.

The object hangs by a metal wire that is connected to earth.
Explain why a metal wire is used to connect the object to earth.

(2)

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(Total for question = 5 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	C lost electrons A is incorrect because it would give the base a negative charge B and D are incorrect because protons do not move in this situation.		(1) AO1

Question number	Answer	Additional guidance	Mark
(ii)	any **two** from electrons have been transferred / moved (1) by friction (1)	cloth has gained electrons accept negative charge for electrons do not credit positive electrons / protons	(2) AO1

Q2.

	Answer	Acceptable answers	Mark

(i)	Correct responses can be seen in (i) or (ii) An explanation linking • electrons (1)and one of • removed by friction (1) • (transferred) to plastic (1)	["positive electrons/ protons moving", seen anywhere in part (i) or (ii) loses this mark] ignore reference to charge before rubbing transferred from cloth	**(2)**
(ii)	opposite to charge on plastic (1) equal to charge on the plastic (1)	charge on cloth is positive same size as charge on plastic electrons transferred from the cloth equal to electrons lost by cloth	**(2)**

Q3.

Question Number:	Answer	Additional Guidance	Mark
(i)	an explanation to link **3** of the following:	reference to positive electrons or positive charge moving loses that mark point	**(3)** AO 2 1
	friction (between cloth and comb) (1)		
	transfer of electrons / charge {from plastic comb / on to the cloth} (1)	electrons/charges are rubbed off comb (on to cloth)	
	electrons carry a negative charge (1)	leaving cloth with negative charge	
	leaving **excess** positive charge on the comb (1)	more protons than electrons (on the comb)	

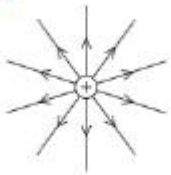
Question Number:	Answer	Additional Guidance	Mark
(ii)	an explanation linking: a negative charge is induced (1)	 allow a clear description of induction ignore references to positive charge being moved in this context only	(3) AO 2 1
	on the part of the paper closest to the comb (1)		
	opposite charges attract (1)	force of attraction sufficient to pick up the pieces of paper	

Q4.

Question number	Answer	Additional guidance	Mark
(i)	An explanation that combines identification - understanding (1 mark) and reasoning - understanding (1 mark): charges move (1)		(2)
	because of **friction** (1)	(negative) electrons transfer glass loses electrons	

Question number	Answer	Mark
(ii)	An explanation that combines identification - understanding (1 mark) and reasoning - understanding (1 mark): (negative) electrons are rubbed off the glass (on to the silk) (1) giving the silk a negative charge (1)	(2)

Q5.

Question Number	Answer	Mark
	A  **The only correct answer is A** ***B*** *is not correct because the arrows are in the wrong direction* ***C*** *is not correct because the field is not circular* ***D*** *is not correct because the field is not circular*	**(1)** AO 1 1

Q6.

	Answer	Additional guidance	Mark
i	field lines / arrows point away from (positive) charge(s) (1)	field lines / arrows show the direction of force(s) on a positive charge accept (because) field lines / arrows point outwards ignore statements about charge flow / current	**(1)** **A01.1**

	Answer	Additional guidance	Mark
ii	field **lines** get further apart / diverge (as x increases) (1)	distance between field **lines** gets bigger concentration of **lines** gets smaller	**(1)** **AO2.1**

	Answer	Additional guidance	Mark
(iii)	rearrangement **and** substitution (1)		**(2)** **AO2.1**

	$A = 20\,000 \times 3^2$ evaluation (1) 180 000 (N cm^2 / C)	 award full marks for the correct answer without working. 1.8 to any other power of 10 scores 1 mark	

Q7.

	Answer	Additional guidance	Mark
	(a region / place / area / volume) where (electric) **charge** experiences / feels a **force** (1)	(a region) where a **charge** experiences attraction / repulsion ignore 'electricity' accept where direction of **force** on a (positive) **charge** (is shown)	**(1)** **A01.1**

Q8.

Question number	Answer	Mark
	The earth wire discharges the aircraft to prevent sparking which could ignite the fuel/cause a fire	**(1)**

Q9.

Question Number	Answer	Acceptable answers	Mark
.(a)(i)	**A** - negative charge has moved from the cloth to the rod		**(1)**

Question Number	Answer	Acceptable answers	Mark
.(a)(ii)	An explanation linking they repelled (1) (strips had) like charge (1)	 push away same (type of) charge	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)(i)	An explanation linking any **two** from charges are separated (1) possibility of a spark (1) ignite the fuel (1)	ignore ref to electric shock pd between plane and ground cause fire / explosion	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)(ii)	An explanation linking **three** from Metals are (good) conductors (1) Electrons/(negative) charge can flow through wire (1) charge goes from/to the ground / earth (1) discharge the tank/aircraft/pipes (1)	 Reject flow of positive charge for this mark plane is earthed/grounded charge does not build up/dissipates Allow no pd between plane and ground so no spark possible for 2 marks	**(3)**

(Total for Question = 8 marks)

Q10.

Question number	Answer	Additional guidance	Mark
(i)	An answer that combines the following points of understanding to provide a logical description: the situation which caused the charge separation (1) where the spark travelled {from/to} (1)	examples: when refuelling, spark between end of {fuel/pipe} and vehicle =2 spark {between/from /to} person comb/clothes/metal handle and, when combing hair/removing clothing/opening door = 2 lightning flash, between cloud and cloud/plane/ground, =2 ignore between plug and socket/jump leads	(2)

Question number	Answer	Additional guidance	Mark
(ii)	unit conversion (1) $0.22\ \mu C = 0.22 \times 10^{-6}$ C and $2\ ms = 2 \times 10^{-3}$ s substitution (1) 0.22×10^{-6} = current x 2×10^{-3} s rearrangement (1) current = $0.22 \times 10^{-6} / 2 \times 10^{-3}$ evaluation (1) 1.1×10^{-4} (A)	substitution and re-arrangement in either order both needed award full marks for correct answer without working power of ten error only loses one mark, if the rest is correct	(4)

Q11.

Question Number	Answer	Mark
(i)	**The only correct answer is A**: It removes electrons from the paint drops **B is incorrect** because that would give an overall negative charge to the drops **C is incorrect** because protons are not transferred **D is incorrect** because protons are not transferred	(1)

Question Number	Answer	Additional Guidance	Mark
(ii)	An explanation linking like charged drops are repelling each other (1) (so) spray is wider (from sprayer Y) (1)	 more spread out / finer / larger area / more dispersed	(2)

Question Number	Answer	Additional Guidance	Mark
(iii)	An explanation linking metal (wire) can conduct electrons / charge (from earth) (1) (so) prevent (positive) charge accumulating on object (1)	 metal is a conductor discharge (the object)	(2)